إسئلة سابقة ..

مختبر الكيمياء

FOLLOW US :

Mech.MuslimEngineer.Net

FB.com/Groups/Mid.Group

0789434018 MechFet

youtube.com/MechanicalFet

لجنة الميكانيك Polytechnic

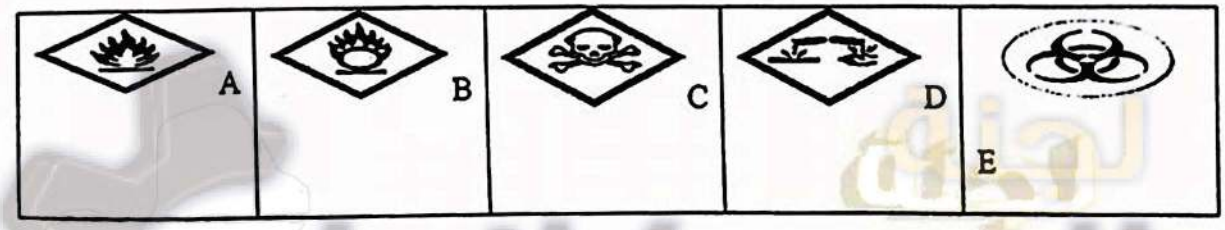
Final Exam 2016

Useful Information	
Atomic masses: (g\mol) Xe:131.3, Cl:35.45, Mg :24.31, S:32, O: 16 , C: 12 ,H:1, Na:23	R = 0.0821 atm. L/K.mol=8.314J\K.mol 1 atm = 101.3 kPa=760mmHg density of vinegar =1.00 g/ml density of bleach = 1.00 g/ml

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
ANSWER																	

art 1 Choose the most correct answer for the following questions and transfer them to the table.(45 r

Q1- The pictogram that correctly indicates an oxidizing chemical hazard is:



Q2- The glassware used to determine the volume of a metal in density experiment is called:

A) Burette B) Test tube C) Crucible D) Desiccator E) graduated cylinder

Q3- A compound was found to contain 55.2% of xenon "Xe" and 44.8% of chlorine "Cl". The empirical formula of this compound is:

A) $XeCl_2$ B) XeCl C) $XeCl_3$ D) $XeCl_4$ E) Xe_2Cl

Q4- For the hydrated salt: $MgSO_4.XH_2O$, if mass of hydrate 2.0 g, and $\%H_2O = 47.3\%$, the value of wate crystallization(X) is:

A) 6 B) 9 C) 15 D) 4 E) 3

Q5- The by-product formed during combustion (burning) of magnesium is :

A) NH_3 B) Mg_3N_2 C) Mg_2N_3 D) H_2O E) MgO

Q6- The temperature (in °C) for a gas that weighs 1.53g (molar mass = 74.0g/mol) and occupies a volu 650.0 mL at pressure of 86.0 kPa equals :

A) 52.5 B) 89.9 C) 14.3 D) 63.1 E)127.7

Q7- During vinegar analysis, a 2.0 ml vinegar sample (% CH_3COOH= 5.5%) was titrated with 0.2 M NaOH, if the initial burette reading was 4.0 mL, then the final burette reading (in mL) equals:

A) 11.17 B) 17.17 C) 13.17 D) 15.17 E) 9.17

Q8- A 30.0 mL of diluted bleach solution was titrated with 19.0 ml of 0.3 M of sodium thiosulfate ($Na_2S_2O_3$). If dilution factor = 5. Then % of NaOCl equals :

A) 3.53% B) 1.41 % C) 4.24 % D) 2.12 % E) 7.07%

Q9- During experiment of vinegar analysis , the color changes from _____ to _____ when end point is reached :

A) deep blue- white(colorless)

B) deep brown- yellow

C) pink- colorless

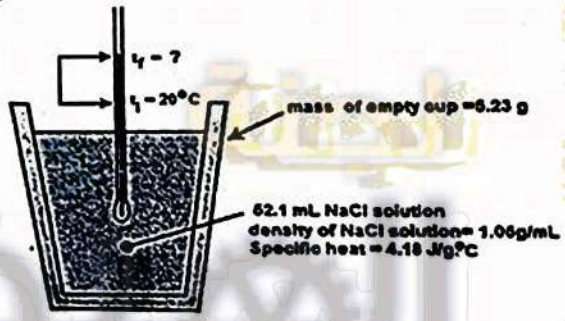
D) Colorless- pink

E) White-deep blue

Q10-The diagram beside shows data concerning experiment of determination of enthalpy of neutralization of the reaction:

$NaOH + HCl \rightarrow NaCl + H_2O$ $\Delta H_{neut} = -71.5$ kJ

Use the data given in the diagram, to find the maximum temperature reached during reaction, if moles of NaOH used in experiment = 0.06 mol



A) 45.0 °C
B) 41.9 °C
C) 48.1 °C
D) 32.5 °C
E) 38.8 °C

Q11- Which of the following statements is incorrect :

A) Endothermic reactions have positive ΔH.

B) Heat evolved by reaction equals heat absorbed by surrounding.

C) $CH_3COO^- \rightarrow H^+ + CH_3COOH$ is a dissociation reaction.

D) $Mg_{(s)} + \frac{1}{2} O_{2(g)} \rightarrow MgO_{(s)}$ is a formation reaction.

E) During acid–base reactions, the cup becomes warmer upon mixing of the acid and base solutions.

Q12- Use the enthalpies of following reactions:

$H_{2(g)} \rightarrow 2H_{(g)}$ $\Delta H = -52.3$ kJ

$2Br_{(g)} \rightarrow Br_{2(g)}$ $\Delta H = 144.5$ kJ

$\frac{1}{2}H_{2(g)} + \frac{1}{2}Br_{2(g)} \rightarrow HBr$ $\Delta H = -210.7$ kJ

To determine the ΔH (in kJ) for the reaction: $H_{(g)} + Br_{(g)} \rightarrow HBr$

A) -107.3 B) -112.3 C) -97.3 D)-102.3 E) -92.3

Q13- Concerning experiment of chemical kinetics, the appearance of the blue color during experiment indicates:

A) Consumption of all I_3^-.

B) Formation of I^- species.

C) Consumption of all $S_2O_3^{2-}$

D) Formation of H_2O_2-starch complex

E) The rate of reaction is zero.

Q14- During experiment of chemical kinetics of the reaction

$H_2O_2 + 2I^- + 2H^+ \rightarrow I_2 + H_2O$

The rate law expression is: $Rate = k [H_2O_2]^1 [I^-]^1$

If rate constant (k) $=3.6 \times 10^{-3}$ and $[H_2O_2] = [I^-] = 0.035M$ and initial concentration of $[S_2O_3^{2-}] = 0.00$

Calculate time (in minutes) required for the blue color to appear?

A) 5.67 B)7.56 C) 9.45 D)15.11 E)11.33

Q15- One of the following does not affect rate of reaction

A) Concentration of reactants
B) Presence of catalyst
C) Chemical nature of reactants
D) Temperature at which reaction occurs
E) Number of reactants participated in chemical reaction

Q16- For a second order reaction, if Arrhenius constant (A) = $1.56X10^{10}$, and activation energy of rea Ea = 67.2 KJ/mol, then the value and unit of rate constant at 40°C equals : (R=8.314 J/m

A) $0.95\ M^{-2}.s^{-1}$ B) $0.0095\ M^{-1}.s^{-1}$ C) $0.95\ M^{-1}.s^{-1}$ D) $0.095\ M^{-1}.s^{-1}$ E) 0.095

Q17- If rate constant $k = 0.0071\ s^{-1}$ at 300 K, and $k = 0.88\ s^{-1}$ at 500 K. Then activation energy of react (kJ /mol) equals:

A) 30.1 B) 18.0 C) 24.0 D) 21.0 E) 19.2

Q18- For the reaction: $3A + 2B + C \rightarrow 2D$ $Rate = k [A]^X[B]^Y[C]^Z$

If the following data were collected for the above reaction

What are the orders of X, Y, Z respectively?

A) 1,3,2
B) 3,2,1
C) 2,1,3
D) 1,2,3
E) 3,1,2

[A]	[B]	[C]	Rate (M/s)
0.23M	0.27M	0.36M	$5.14X10^{-3}$
0.23M	0.27M	0.12M	$5.71X10^{-3}$
0.92M	0.27M	0.36M	0.206
0.23M	0.54M	0.12M	0.046

Part 2 Fill the blanks with the correct answer of the following (increase ,decrease ,no effect) (5 marks)

1- Incomplete dehydration of a hydrated salt , will cause the value of water of crystallization (X) to	
2- During burning of magnesium turnings, if magnesium was not allowed to react completely with oxygen, then ratio of Mg/O in the empirical value will	
3- If a sample of 1.0 ml of concentrated vinegar was diluted by adding 10.0 mL of water instead of 5.0 mL water, then molarity of diluted vinegar solution will	
4- Dilution of a bleach sample before titration with $Na_2S_2O_3$, will cause the volume of $Na_2S_2O_3$ required to reach the end point to	
5- When external atmospheric pressure increases, the boiling point of a certain liquid is	

Al-Balqa Applied University

Faculty of Engineering Technology, Department of Applied Sciences

General Chemistry Laboratory (CHEM 111) Final Exam

Date: 1 /8/2011

Student name: ----------

Registration No: ----------

Instructor Name: ----------

Time of Lab: ----------

Helpful data: $R = 0.0821\ L.atm.mol^{-1}.K^{-1} = 8.314\ J.mol^{-1}.K^{-1}$, $K = 273 + °C$, specific heat $H_2O = 4.184\ J.g^{-1}.°C^{-1}$, $\ln(K_2/K_1) = -E_a/R\ (1/T_2 - 1/T_1)$ where $T_2 > T_1$, 1 atom = 760 mmHg,

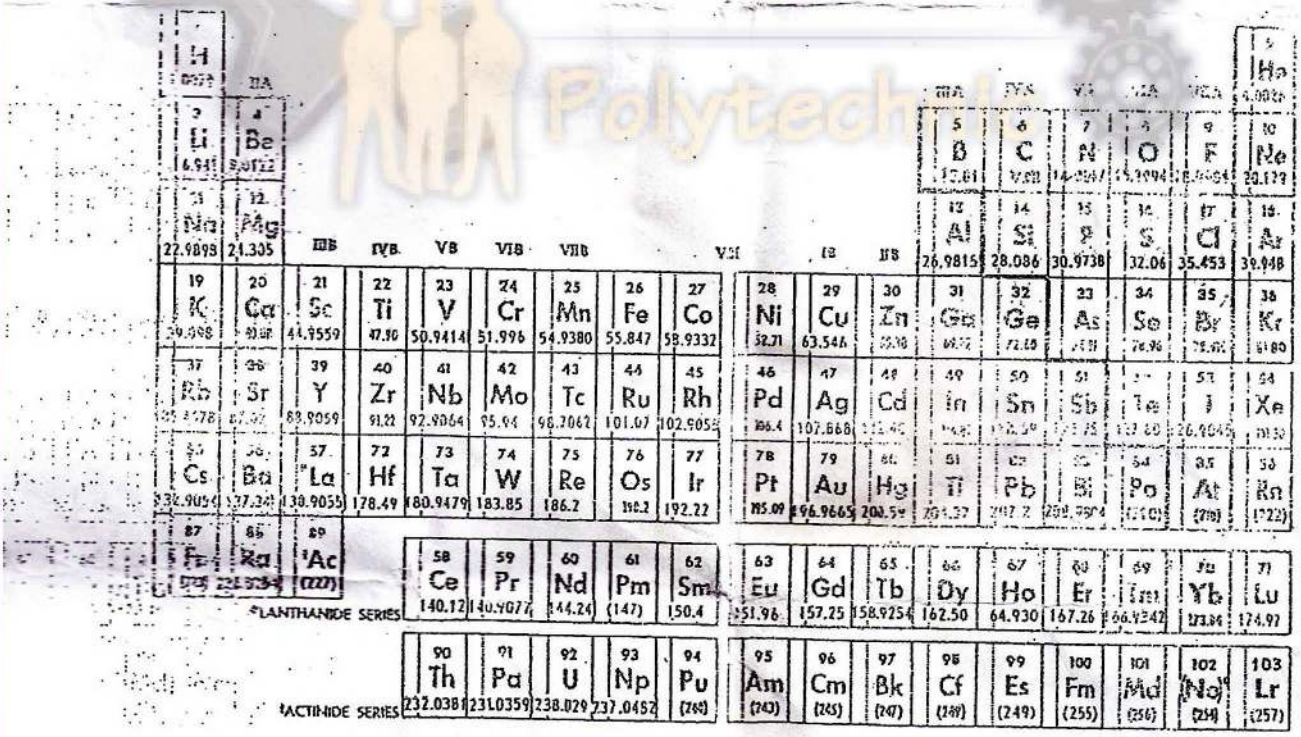
Transfer all your answer to the table below

Answer Sheet

(E)

Question No	1	2	3	4	5	6	7	8	9	10	11	12	13
Answer	D	B	D	A	C	E	D	A	E	C	a	B	e

Question No	14	15	16	17	18	19	20	21	22	23	24	25	26
Answer	C	B	D	B			e						



Choose the most correct answer for each of the following questions:

Safety and physical properties

1. Circle the most correct statement:
   a) Smoking, eating, and drinking are allowed in the laboratory.
   b) Always pour water into acids.
   c) Taste the chemical in order to identify it.
   d) Don't heat flammable liquids over direct flame.
   e) Smell directly the chemicals in order to identify them.

2. Circle the correct statement:
   a) Solubility is the amount of solute dissolved in a certain amount of solvent at a given temperature to give unsaturated solution.
   b) Boiling point is the point at which the vapor pressure of the substance equals the external pressure.
   c) Density describes the volume per unit mass.
   d) Boiling point determines the purity of a substance
   e) Melting point is independent on the external pressure.

3. Which equipment is used to measure the temperature of a liquid?
   a) balance b) graduated cylinder c) desiccator d) thermometer
   e) Bunsen burner

4. If the density of La is 6.17 g/cm^3, then the volume (in cm^3) of 0.22 kg La equals:
   a) 35.66 b) 39.90 c) 11.40 d) 30.57 e) 17.10

5. In an experiment, NaCl dissolves in a certain solvent, this means that the solvent is:
   a) Nonpolar b) has high density c) polar d) has low B.P e) has low density

Water of crystallization

6. The % by mass of H_2O in a hydrated salt is 46.8 % and X in this salt equals 6. The molecular weight (in g/mol) of this hydrated salt equals:
   a) 474 b) 249.6 c) 278.1 d) 225 e) 230.8

Empirical formula

7. A chemical compound contains C and H. If the % by mass of C is 75.0%, then the empirical formula of the compound is:
   a) CH_3 b) C_2H_5 c) CH_2 d) CH_4 e) C_3H_7

8. One of the following is not an empirical formula:
   a) HOOCCOOH b) CH_3COOCH_3 c) CH_3CH_2COOH d) CH_3COCH_3 e) CH_3OH

Volatile liquid

9. The volume of 14.0 g of $C_4H_{8(g)}$ at 80 °C and 0.9 atm equals:
   a) 3.2 L b) 32 mL c) 0.32 mL d) 6.4 L e) 8.05 L

10. The molecular weight of one of the following compounds can be determined by dumas method employed for volatile liquids:
    a) C_9H_{20} (B.P = 151°C) b) $C_{10}H_{22}$ (B.P = 174°C)
    c) C_6H_{14} (B.P = 68°C) d) C_7H_{16} (B.P = 100°C) e) $C_{11}H_{24}$ (B.P = 184°C)

Volumetric analysis I and II

11. The % (m/m) of CH_3COOH in commercial vinegar is 2% (density = 1.01 g/mL). The volume (in mL) of 0.1 M NaOH solution needed to completely neutralize the acid in 10.0 mL sample of this vinegar equals:
a) 33.67 b) 67.33 c) 84.17 d) 117.8 e) 50.50

12. Starch solution was used in bleach analysis as:
a) buffer b) an indicator c) an acid d) a catalyst e) solvent

13. During redox reaction experiment, 20.0 mL of 0.102 M $S_2O_3^{2-}$ solution was needed to reduce I_3^- produced from the oxidation of I^- with 4.0 mL of bleach solution, the molar concentration of NaOCl in the bleach solution equals:
a) 0.150 b) 0.300 c) 0.521 d) 0.24 e) 0.255

14. In the redox reaction $S_2O_3^{2-}{}_{(aq)} + I_3^-{}_{(aq)} \rightarrow 3I^-{}_{(aq)} + S_4O_6^{2-}{}_{(aq)}$, the oxidizing agent is :
a) S b) O c) I_3^- d) I^- e) $S_2O_3^{2-}$

15. In the reaction $I^-{}_{(aq)} + H_2O_{2(aq)} \rightarrow I_3^-{}_{(aq)} + H_2O_{(l)}$ in acidic solution, the number of electrons gained by four mole of H_2O_2 equals ------ electrons:
a) 2 b) 8 c) 6 d) 1 e) 4

16. During the analysis of vinegar, NaOH solution was filled in an instrument called:
a) desicator b) beaker c) crucible d) burete e) flask

Thermodynamics I and II

17. given the following thermo-chemical equations

$GeO_{(s)} + \frac{1}{2} O_{2(g)} \rightarrow GeO_{2(s)}$ $\Delta H° = -279$ kJ
$Ge_{(s)} + O_{2(g)} \rightarrow GeO_{2(s)}$ $\Delta H° = -534$ kJ
$\Delta H°_f$ for $GeO_{(s)}$ equals:

a) + 225 b) – 255 c) – 0.225 d) 25.5 e) – 25.5

18. For the reaction $2CH_3CH_2OH + 6O_{2(g)} \rightarrow 4CO_{2(g)} + 6H_2O_{(l)}$ $\Delta H = -1199$ kJ. The mass (in grams) of CH_3CH_2OH needed to produce 2359 kJ of heat energy equals:
a) 303.6 b) 274.6 c) 414.1 d) 150.5 e) 276

19. Consider the reaction $MgO_{(s)} + HCl_{(aq)} \rightarrow MgCl_{2(aq)} + H_2O_{(l)}$. When 0.4 g of MgO was mixed with 25.0 ml of 0.5 M HCl solution the temperature rose from 22 °C to 36.5 °C , if mass of $MgCl_2$ solution was 25.4 g then enthalpy change for this reaction (in kJ/mol MgO) equals:
a) 181 b) -181 c) -146.5 d) 221 e) - 145.5

20. One of the following equations represents an endothermic reaction:
a) $Mg_{(s)} + \frac{1}{2} O_{2(g)} \rightarrow MgO_{(s)}$ b) $MgO_{(s)} + HCl_{(aq)} \rightarrow MgCl_{2(aq)} + H_2O_{(l)}$
c) $OH^-{}_{(aq)} + H^+{}_{(aq)} \rightarrow H_2O_{(l)}$ d) $Mg_{(s)} + HCl_{(aq)} \rightarrow MgCl_{2(aq)} + H_{2(g)}$
e) $CH_3COOH_{(l)} \rightarrow CH_3COO^-{}_{(aq)} + H^+{}_{(aq)}$

Chemical kinetics

21. For the reaction, $H_2O_2 + 2I^- + 2H^+ \rightarrow I_2 + 2H_2O$, Where H^+ represents the buffer solution: Which of the following statements is correct?

a) The rate of consumption of H_2O_2 = twice the rate of formation of H_2O

b) The reaction is first order with respect to H^+

c) The time required for the blue color to appear indicates that the concentration of sodium thiosulfate is zero

d) The blue color that appears is due to formation of a phenolphthalein-starch complex.

e) The end point is reached when the color turns from blue to colorless.

22. During the chemical kinetic experiment, the concentration of $[H^+]$ was fixed constant by using:

a) $Na_2S_2O_3$ b) buffer c) $H_2O_{2(aq)}$ d) CH_3COOH e) $HCl_{(aq)}$

23. For the reaction $I^-_{(aq)} + H_2O_{2(aq)} \rightarrow I_3^-{}_{(aq)} + H_2O_{(l)}$, the rate of the reaction was measured as:

a) $\frac{1}{2} \Delta [I_3^-] / t$ b) $\Delta[S_2O_3^{2-}]_f / t$ c) $\frac{1}{2} \Delta [I^-] / t$ d) $\frac{1}{2} \Delta[S_2O_3^{2-}]_f / t$ e) $\frac{1}{2} \Delta[I_3^-] / t$

24. During the chemical kinetics experiment for the reaction $H_2O_2 + 2I^- + 2H^+ \rightarrow I_2 + 2H_2O$, a student mixes the following solutions with their initial concentrations as shown in the table:

$[KI]_i$	$[H_2O_2]_i$	$[S_2O_3^{2-}]_i$	starch	Time needed for run (min)
0.3	0.2	0.09	1 mL	3

The rate of reaction (in M/s) equals:

a) 1.94×10^{-4} b) 2.50×10^{-4} c) 1.38×10^{-4} d) 1.67×10^{-4} e) 1.11×10^{-4}

25. The following data were collected for the reaction: $OCl^- + I^- \rightarrow OI^- + Cl^-$

Initial concentration (mol/L)		Rate of reaction ($mol\ L^{-1}\ s^{-1}$)
$[OCl^-]$	$[I^-]$	
1.7×10^{-3}	1.7×10^{-3}	1.75×10^{-4}
3.4×10^{-3}	1.7×10^{-3}	3.50×10^{-4}
1.7×10^{-3}	3.4×10^{-3}	3.50×10^{-4}

The overall order of this reaction is:

a) 1'st b) 5'th c) 3'rd d) 4'th e) 2'nd

26. For the reaction $4NH_{3(g)} + 3O_{2(g)} \rightarrow 2N_{2(g)} + 6H_2O_{(g)}$, If the rate of formation of N_2 was 0.68 $M.\ s^{-1}$, then the rate of O_2 consumption (in $M.\ s^{-1}$) equals:

a) 1.02 b) 2.04 c) 0.453 d) 0.227 e) 1.36

